Отчет по практической работе №1

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Группа: [БСБО-16-23]

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1. Выполненные команды Docker

1.1 Работа с образами

Поиск образов в Docker Hub

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker search nginx
NAME                                      DESCRIPTION                                     STARS     OFFICIAL
nginx                                     Official build of Nginx.                        21194     [OK]
nginx/nginx-ingress                       NGINX and  NGINX Plus Ingress Controllers fo…   115
nginx/nginx-prometheus-exporter           NGINX Prometheus Exporter for NGINX and NGIN…   50
nginx/unit                                This repository is retired, use the Docker o…   66
nginx/nginx-ingress-operator              NGINX Ingress Operator for NGINX and NGINX P…   3
nginx/nginx-quic-qns                      NGINX QUIC interop                              1
nginx/nginxaas-loadbalancer-kubernetes                                                    1
nginx/unit-preview                        Unit preview features                           0
bitnami/nginx                             Bitnami Secure Image for nginx                  203
bitnamicharts/nginx                       Bitnami Helm chart for NGINX Open Source        3
ubuntu/nginx                              Nginx, a high-performance reverse proxy & we…   139
kasmweb/nginx                             An Nginx image based off nginx:alpine and in…   8
rancher/nginx                                                                             3
linuxserver/nginx                         An Nginx container, brought to you by LinuxS…   236
dtagdevsec/nginx                          T-Pot Nginx                                     0
vmware/nginx                                                                              3
paketobuildpacks/nginx                                                                    0
chainguard/nginx                          Build, ship and run secure software with Cha…   5
gluufederation/nginx                       A customized NGINX image containing a consu…   1
cleanstart/nginx                          Secure by Design, Built for Speed, Hardened …   0
antrea/nginx                              Nginx server used for Antrea e2e testing        0
intel/nginx                                                                               0
circleci/nginx                            This image is for internal use                  2
docksal/nginx                             Nginx service image for Docksal                 1
activestate/nginx                         ActiveState's customizable, low-to-no vulner…   0

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ |
```

Скачивание образа

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker pull nginx:alpine
alpine: Pulling from library/nginx
589002ba0eae: Pull complete
bca5d04786e1: Pull complete
3e2c181db1b0: Pull complete
6b7b6c7061b7: Pull complete
399d0898a94e: Pull complete
955a8478f9ac: Pull complete
6d397a54a185: Pull complete
5e7756927bef: Pull complete
Digest: sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca557a8d7702ba
Status: Downloaded newer image for nginx:alpine
docker.io/library/nginx:alpine

What's next:
    View a summary of image vulnerabilities and recommendations → docker scout quickview nginx:alpine

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ |
```

Просмотр локальных образов

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker images
                                                                                    i Info →   U  In Use
IMAGE                                  ID             DISK USAGE   CONTENT SIZE   EXTRA
devprom/alm-app:latest                 e761777db47f       1.75GB             0B
devprom/math:latest                    123cb19d61b0        970MB             0B
jgraph/drawio:latest                   cf96a61257e9        881MB             0B
mariadb:10.7                           895b6c8829c3        396MB             0B
nginx:alpine                           b76de378d572       62.1MB             0B
plantuml/plantuml-server:latest        85b1c65e78ae        525MB             0B
postgres:16.3                          07a4ee949b9e        432MB             0B

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$
```

Просмотр истории слоев образа

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker history nginx:alpine
IMAGE          CREATED       CREATED BY                                      SIZE      COMMENT
b76de378d572   3 weeks ago   RUN /bin/sh -c set -x     && apkArch="$(cat …   49.4MB    buildkit.dockerfile.v0
<missing>      3 weeks ago   ENV ACME_VERSION=0.3.1                          0B        buildkit.dockerfile.v0
<missing>      3 weeks ago   ENV NJS_RELEASE=1                               0B        buildkit.dockerfile.v0
<missing>      3 weeks ago   ENV NJS_VERSION=0.9.5                           0B        buildkit.dockerfile.v0
<missing>      3 weeks ago   CMD ["nginx" "-g" "daemon off;"]                0B        buildkit.dockerfile.v0
<missing>      3 weeks ago   STOPSIGNAL SIGQUIT                              0B        buildkit.dockerfile.v0
<missing>      3 weeks ago   EXPOSE map[80/tcp:{}]                           0B        buildkit.dockerfile.v0
<missing>      3 weeks ago   ENTRYPOINT ["/docker-entrypoint.sh"]            0B        buildkit.dockerfile.v0
<missing>      3 weeks ago   COPY 30-tune-worker-processes.sh /docker-ent…   4.62kB    buildkit.dockerfile.v0
<missing>      3 weeks ago   COPY 20-envsubst-on-templates.sh /docker-ent…   3.02kB    buildkit.dockerfile.v0
<missing>      3 weeks ago   COPY 15-local-resolvers.envsh /docker-entryp…   389B      buildkit.dockerfile.v0
<missing>      3 weeks ago   COPY 10-listen-on-ipv6-by-default.sh /docker…   2.12kB    buildkit.dockerfile.v0
<missing>      3 weeks ago   COPY docker-entrypoint.sh / # buildkit          1.62kB    buildkit.dockerfile.v0
<missing>      3 weeks ago   RUN /bin/sh -c set -x     && addgroup -g 101…   4.25MB    buildkit.dockerfile.v0
<missing>      3 weeks ago   ENV DYNPKG_RELEASE=1                            0B        buildkit.dockerfile.v0
<missing>      3 weeks ago   ENV PKG_RELEASE=1                               0B        buildkit.dockerfile.v0
<missing>      3 weeks ago   ENV NGINX_VERSION=1.29.5                        0B        buildkit.dockerfile.v0
<missing>      3 weeks ago   LABEL maintainer=NGINX Docker Maintainers <d…   0B        buildkit.dockerfile.v0
<missing>      4 weeks ago   CMD ["/bin/sh"]                                 0B        buildkit.dockerfile.v0
<missing>      4 weeks ago   ADD alpine-minirootfs-3.23.3-x86_64.tar.gz /…   8.44MB    buildkit.dockerfile.v0

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$
```

Удаление образа

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker rmi nginx:alpine
Untagged: nginx:alpine
Untagged: nginx@sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca557a8d7702ba
Deleted: sha256:b76de378d57272a1dd9091a05dd548a3639dfb792ebdbf95d06704d2950afdea
Deleted: sha256:278162406e881ff08253c6eff4802ae0c7eecacc5cf442edfd2648aff5ad4497
Deleted: sha256:3cc477c2741995382b41fc38f3c217bb57462b82a92a00fcba37d3d6d9d009b2
Deleted: sha256:a74be0c2915d6076b248d155341d2c187217669b39754b2d1cbc305407f1fec5
Deleted: sha256:e174ca5cc17551940f3aae50dcf30902076216f95020109acf2a5e6d9f926ce5
Deleted: sha256:28936064555d92e3ecf96d4085e548664795f5469a90042d0c5a9742117fe939
Deleted: sha256:ca8b88624ae0323b7f59f4fb8a8c0977d96f44e94bae4b37df53d865a6c49fd1
Deleted: sha256:71bd4a78d25ba3d319bbea9f2341b6376152a468be81c26ee26fe94169b56ec3
Deleted: sha256:989e799e634906e94dc9a5ee2ee26fc92ad260522990f26e707861a5f52bf64e

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$
```

Практическое задание

Поиск образа postgres:15-alpine

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker search postgres:15-alpine
NAME                      DESCRIPTION                                     STARS     OFFICIAL
postgres                  The PostgreSQL object-relational database sy…   14805     [OK]
cimg/postgres                                                             9
circleci/postgres         The PostgreSQL object-relational database sy…   35
kasmweb/postgres          Postgres image maintained by Kasm Technologi…   6
elestio/postgres          Postgres, verified and packaged by Elestio      2
ubuntu/postgres           PostgreSQL is an open source object-relation…   42
chainguard/postgres       Build, ship and run secure software with Cha…   1
artifacthub/postgres                                                      0
cleanstart/postgres       Secure by Design, Built for Speed, Hardened …   0
corpusops/postgres        https://github.com/corpusops/docker-images/     0
geokrety/postgres         Postgres with postgis + quantile and amqp ex…   0
rootpublic/postgres                                                       0
dockette/postgres         My PostgreSQL image with tunning and preinst…   1
vulhub/postgres                                                           1
wayofdev/postgres                                                         0
uselagoon/postgres                                                        0
openeuler/postgres                                                        0
pgrouting/postgres         Postgres Docker images with PostGIS and dep…   0
clarinpl/postgres                                                         0
supabase/postgres         Unmodified Postgres with some useful plugins…   73
mosipdev/postgres                                                         0
brimstone/postgres        postgres image with traefik-cert support        0
trainlineeurope/postgres  Extended version of official Postgres https:…   0
blacklabelops/postgres    Postgres Image for Atlassian Applications       4
fredboat/postgres         PostgreSQL 10.0 used in FredBoat's docker-co…   1

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ |
```

Скачивание образа postgres:15-alpine

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker pull postgres:15-alpine
15-alpine: Pulling from library/postgres
589002ba0eae: Already exists
aa104f8e5799: Pull complete
ac1ff2379be8: Pull complete
8c9e2aa22b4c: Pull complete
765f80b9a483: Pull complete
a27cb555d458: Pull complete
610258b635a8: Pull complete
2a9c4fd442cb: Pull complete
9f6d6f38d9d3: Pull complete
b53228f7876a: Pull complete
34c88efd36aa: Pull complete
Digest: sha256:2bf082886c059b4cf5af3f9e8d3a8916c35699062899061c13bc32d5921c631a
Status: Downloaded newer image for postgres:15-alpine
docker.io/library/postgres:15-alpine

What's next:
    View a summary of image vulnerabilities and recommendations → docker scout quickview postgres:15-alpine

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ |
```

Поиск образа golang:1.21-alpine

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker search golang:1.21-alpine
NAME                      DESCRIPTION                                     STARS     OFFICIAL
golang                    Go (golang) is a general purpose, higher-lev…   5106      [OK]
circleci/golang           CircleCI images for Go                          21
bitnami/golang            Bitnami Secure Image for golang                 12
okteto/golang             Development environment for golang              6
antrea/golang             Golang image used by Antrea for building        3
rancher/golang                                                            0
alpinelinux/golang        Build container for golang based on Alpine L…   4
corpusops/golang          https://github.com/corpusops/docker-images/     0
rootpublic/golang                                                         0
goharbor/golang                                                           0
jetbrainsinfra/golang     Golang + custom build tools                     2
pivotalcfreleng/golang    this image is deprecated. Please use gcr.io/…   1
jelastic/golang           An image of the Golang application server ma…   0
webhippie/golang          Docker image for Golang                         1
odpks/golang                                                              0
idoall/golang             golang                                          0
amd64/golang              Go (golang) is a general purpose, higher-lev…   3
arm64v8/golang            Go (golang) is a general purpose, higher-lev…   19
arm32v7/golang            Go (golang) is a general purpose, higher-lev…   26
winamd64/golang           Go (golang) is a general purpose, higher-lev…   1
i386/golang               Go (golang) is a general purpose, higher-lev…   2
s390x/golang              Go (golang) is a general purpose, higher-lev…   0
cfinfrastructure/golang                                                   0
cycloid/golang                                                            0
ppc64le/golang            Go (golang) is a general purpose, higher-lev…   1

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$
```

Скачивание образа golang:1.21-alpine

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker pull golang:1.21-alpine
1.21-alpine: Pulling from library/golang
c6a83fedfae6: Pull complete
41db7493d1c6: Pull complete
54bf7053e2d9: Pull complete
4579008f8500: Pull complete
4f4fb700ef54: Pull complete
Digest: sha256:2414035b086e3c42b99654c8b26e6f5b1b1598080d65fd03c7f499552ff4dc94
Status: Downloaded newer image for golang:1.21-alpine
docker.io/library/golang:1.21-alpine

What's next:
    View a summary of image vulnerabilities and recommendations → docker scout quickview golang:1.21-alpine

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ |
```

Просмотр образов

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker images
                                                                  i Info →   U  In Use
IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
devprom/alm-app:latest               e761777db47f       1.75GB             0B
devprom/math:latest                  123cb19d61b0        970MB             0B
golang:1.21-alpine                   c2321c7cf721        222MB             0B
jgraph/drawio:latest                 cf96a61257e9        881MB             0B
mariadb:10.7                         895b6c8829c3        396MB             0B
nginx:alpine                         b76de378d572       62.1MB             0B
plantuml/plantuml-server:latest      85b1c65e78ae        525MB             0B
postgres:15-alpine                   cd848ee12e8e        274MB             0B
postgres:16.3                        07a4ee949b9e        432MB             0B

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ |
```

1.2 Работа с контейнерами

Запуск контейнера alpine в интерактивном режиме и выполнение команд внутри контейнера

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker run -it --name test-alpine alpine:latest sh
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
589002ba0eae: Already exists
Digest: sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659
Status: Downloaded newer image for alpine:latest
/ # ls -la
total 64
drwxr-xr-x    1 root     root          4096 Feb 28 12:36 .
drwxr-xr-x    1 root     root          4096 Feb 28 12:36 ..
-rwxr-xr-x    1 root     root             0 Feb 28 12:36 .dockerenv
drwxr-xr-x    2 root     root          4096 Jan 27 21:19 bin
drwxr-xr-x    5 root     root           360 Feb 28 12:36 dev
drwxr-xr-x    1 root     root          4096 Feb 28 12:36 etc
drwxr-xr-x    2 root     root          4096 Jan 27 21:19 home
drwxr-xr-x    6 root     root          4096 Jan 27 21:19 lib
drwxr-xr-x    5 root     root          4096 Jan 27 21:19 media
drwxr-xr-x    2 root     root          4096 Jan 27 21:19 mnt
drwxr-xr-x    2 root     root          4096 Jan 27 21:19 opt
dr-xr-xr-x  294 root     root             0 Feb 28 12:36 proc
drwx------    1 root     root          4096 Feb 28 12:37 root
drwxr-xr-x    3 root     root          4096 Jan 27 21:19 run
drwxr-xr-x    2 root     root          4096 Jan 27 21:19 sbin
drwxr-xr-x    2 root     root          4096 Jan 27 21:19 srv
dr-xr-xr-x   13 root     root             0 Feb 28 12:36 sys
drwxrwxrwt    2 root     root          4096 Jan 27 21:19 tmp
drwxr-xr-x    7 root     root          4096 Jan 27 21:19 usr
drwxr-xr-x   11 root     root          4096 Jan 27 21:19 var
/ # exit

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ |
```

Запуск контейнера в фоновом режиме

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker run -d --name web-server -p 8080:80 nginx:alpine
da9ad95458dd97debe782949d6c3b6eaef1e183ceb521290767134c98975377f

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$
```

Просмотр запущенных контейнеров

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                                       NAMES
da9ad95458dd   nginx:alpine   "/docker-entrypoint.…"   25 seconds ago   Up 24 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp     web-server

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$
```

Просмотр всех контейнеров (включая остановленные)

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED              STATUS                     PORTS                                       NAMES
da9ad95458dd   nginx:alpine   "/docker-entrypoint.…"   About a minute ago   Up About a minute          0.0.0.0:8080->80/tcp, [::]:8080->80/tcp     web-server
138e6fc887f5   alpine:latest  "sh"                     5 minutes ago        Exited (0) 3 minutes ago                                               test-alpine

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ |
```

Просмотр логов контейнера

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker logs web-server
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/02/28 12:40:07 [notice] 1#1: using the "epoll" event method
2026/02/28 12:40:07 [notice] 1#1: nginx/1.29.5
2026/02/28 12:40:07 [notice] 1#1: built by gcc 15.2.0 (Alpine 15.2.0)
2026/02/28 12:40:07 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/02/28 12:40:07 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/02/28 12:40:07 [notice] 1#1: start worker processes
2026/02/28 12:40:07 [notice] 1#1: start worker process 30
2026/02/28 12:40:07 [notice] 1#1: start worker process 31
2026/02/28 12:40:07 [notice] 1#1: start worker process 32
2026/02/28 12:40:07 [notice] 1#1: start worker process 33
2026/02/28 12:40:07 [notice] 1#1: start worker process 34
2026/02/28 12:40:07 [notice] 1#1: start worker process 35
2026/02/28 12:40:07 [notice] 1#1: start worker process 36
2026/02/28 12:40:07 [notice] 1#1: start worker process 37
2026/02/28 12:40:07 [notice] 1#1: start worker process 38
2026/02/28 12:40:07 [notice] 1#1: start worker process 39
2026/02/28 12:40:07 [notice] 1#1: start worker process 40
2026/02/28 12:40:07 [notice] 1#1: start worker process 41

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$
```

Подключение к работающему контейнеру

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker exec -it web-server sh
/ # cat /usr/share/nginx/html/index.html
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
/ # exit

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug web-server
    Learn more at https://docs.docker.com/go/debug-cli/

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$
```

Остановка контейнера

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker stop web-server
web-server

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ |
```

Запуск остановленного контейнера

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker start web-server
web-server

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$
```

Удаление контейнера

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker rm web-server
web-server

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ |
```

Практическое задание

Запуск контейнера с PostgreSQL и выполнение команды psql в контейнере

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker run -d --name postgres-15-test -e POSTGRES_PASSWORD=qazxsw512 postgres:15-alpine
3f16873bfb98458126a3ede8da545c749a07e603dfe276cc14582a54b0d48e3d

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker exec -it postgres-15-test sh
/ # psql -U postgres -c "SELECT version();"
                                        version
-----------------------------------------------------------------------------------------
 PostgreSQL 15.17 on x86_64-pc-linux-musl, compiled by gcc (Alpine 15.2.0) 15.2.0, 64-bit
(1 row)

/ # |
```

1.3 Работа с томами

Создание именованного тома

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker volume create my-app-data
my-app-data

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$
```

Просмотр томов

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker volume ls
DRIVER    VOLUME NAME
local     1a1bab0805b3319773f2dd71dedd7f73f618c0707e03eb522a039d6c63d60b61
local     05400298e3c642f2d3d6c72d7d2f99f03337ad0d8ef21cb6a20285ace8c109ed
local     docker_dbdata
local     docker_dbdata_pgsql
local     ec3bf875afadeaac8b76b211189a951b8347e48f0fcd6d08657a99f1453a0568
local     my-app-data

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ |
```

Информация о томе

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker volume inspect my-app-data
[
    {
        "CreatedAt": "2026-02-28T13:13:35Z",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/my-app-data/_data",
        "Name": "my-app-data",
        "Options": null,
        "Scope": "local"
    }
]

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ |
```

Запуск контейнера с томом

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker run -d --name postgres-db -e POSTGRES_PASSWORD=secret -e POSTGRES_DB=testdb -v my-app-data:/var/lib/postgresql/data -p 5432:5432 postgres:15-alpine
8b7fe508b4269979b71557984aafa01ad23083f4e7e582490758ab0e1edc7f80

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$
```

Создание тестовой таблицы

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker exec -it postgres-db psql -U postgres -d testdb -c "CREATE TABLE users (id
SERIAL, name TEXT);"
CREATE TABLE

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug postgres-db
    Learn more at https://docs.docker.com/go/debug-cli/

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ |
```

Занесение данных в таблицу

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker exec -it postgres-db psql -U postgres -d testdb -c "INSERT INTO users
(name) VALUES ('SSA 16');"
INSERT 0 1

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug postgres-db
    Learn more at https://docs.docker.com/go/debug-cli/

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$
```

Остановка и удаление контейнера (данные сохранятся в томе)

```
giper@LAPTOP-31PBTHTU MINGW64 ~
$ docker stop postgres-db
postgres-db

giper@LAPTOP-31PBTHTU MINGW64 ~
$ docker rm postgres-db
postgres-db

giper@LAPTOP-31PBTHTU MINGW64 ~
$
```

Запуск нового контейнера с тем же томом

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker run -d --name new-postgres -e POSTGRES_PASSWORD=secret -e POSTGRES_DB=testdb -v my-app-data:/var/lib/postgresql/data -p 5432:5432 postgres:15-alpine
440a08ff267de6e02bbaaf794f4a8bbfd6b39ce0ad74292f8eda2aa357a9c16c

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ |
```

Проверка, что данные сохранились

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker exec -it new-postgres psql -U postgres -d testdb -c "SELECT * FROM users;"
 id |  name
----+--------
  1 | SSA 16
(1 row)


What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug new-postgres
    Learn more at https://docs.docker.com/go/debug-cli/

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$
```

Практическое задание

Создание тома для статических файлов

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker volume create my-nginx
my-nginx

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$
```

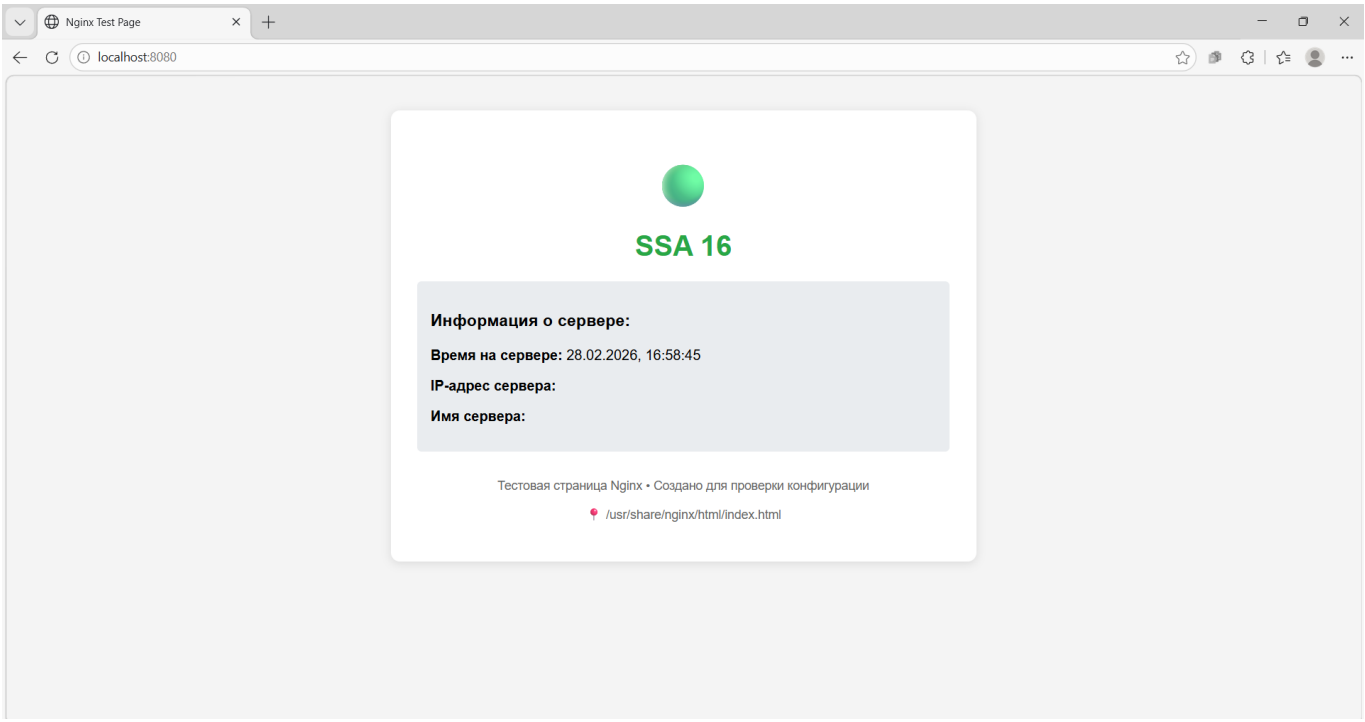
Запуск контейнера с nginx

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker run -d --name my-nginx-server -v my-nginx:/usr/share/nginx/html -p 8080:80 nginx:alpine
5d7c48d7ce6a12867a5da35308c2994b1600f63f07384318935d6b1fc99c85ca

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ |
```

Копирование файла index.html внутрь контейнера

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker cp index.html my-nginx-server:/usr/share/nginx/html
Successfully copied 3.07kB to my-nginx-server:/usr/share/nginx/html

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$
```

Страница в браузере



Сеть в Docker

Создание сети

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker network create my-app-network
d464d2fffb3638b0b8cbc63490d9b888fe5d205b237eb800b24a678c1002e74e

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ |
```

Просмотр сетей

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker network ls
NETWORK ID     NAME            DRIVER    SCOPE
f3e1627293d8   bridge          bridge    local
3855c3534961   host            host      local
d464d2fffb36   my-app-network  bridge    local
588882601faf   none            null      local

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$
```

Запуск контейнеров в одной сети

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker run -d --name postgres --network my-app-network -e POSTGRES_PASSWORD=secret -e POSTGRES_DB=myapp postgres:15-alpine
a4ae32bc378aad806f5aa399e34cf506b06b3a8e297af1d954dde6586b6539e7

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker run -d --name app --network my-app-network -p 8080:80 nginx:alpine
fd74516c458f67962ec4cc3a6917da76601937cb7a3ed7f00e6df24338661f50

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$
```

Проверка связи между контейнерами

```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker exec app ping postgres
PING postgres (172.18.0.2): 56 data bytes
64 bytes from 172.18.0.2: seq=0 ttl=64 time=0.795 ms
64 bytes from 172.18.0.2: seq=1 ttl=64 time=0.122 ms
64 bytes from 172.18.0.2: seq=2 ttl=64 time=0.213 ms
64 bytes from 172.18.0.2: seq=3 ttl=64 time=0.228 ms
64 bytes from 172.18.0.2: seq=4 ttl=64 time=0.219 ms
```

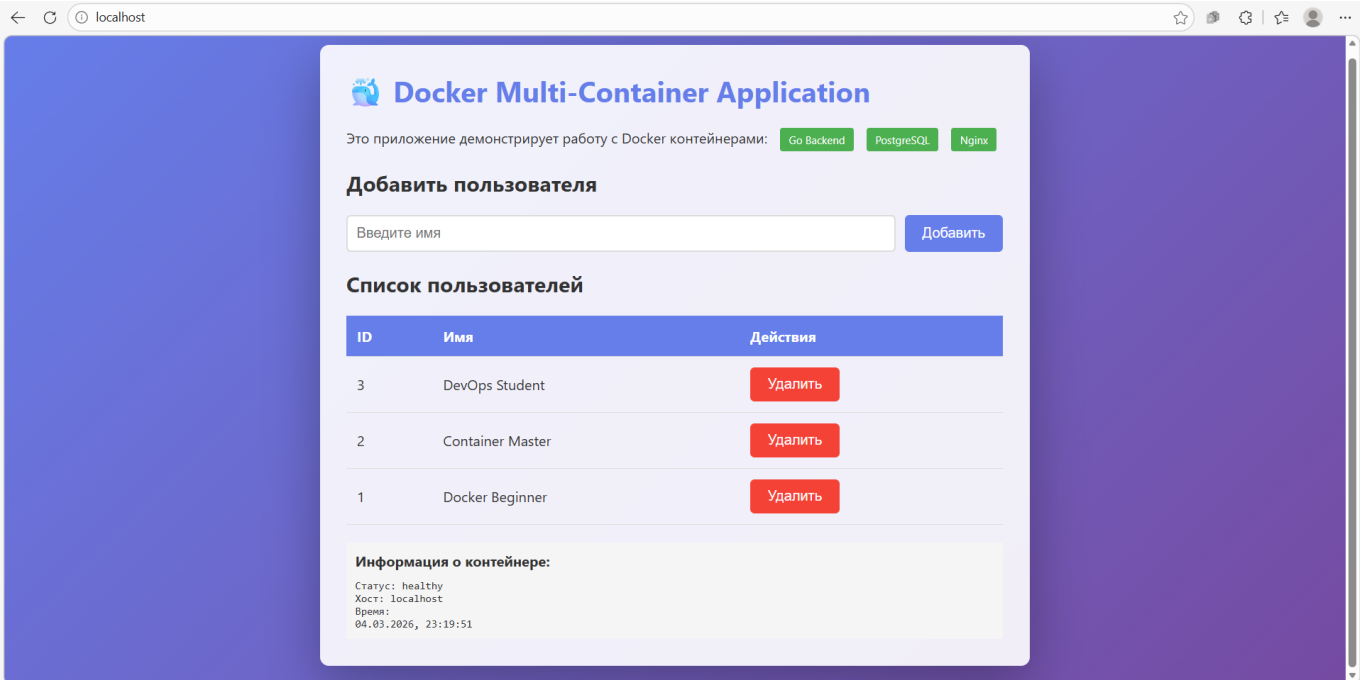
Практическое задание

Взаимодействие между контейнерами

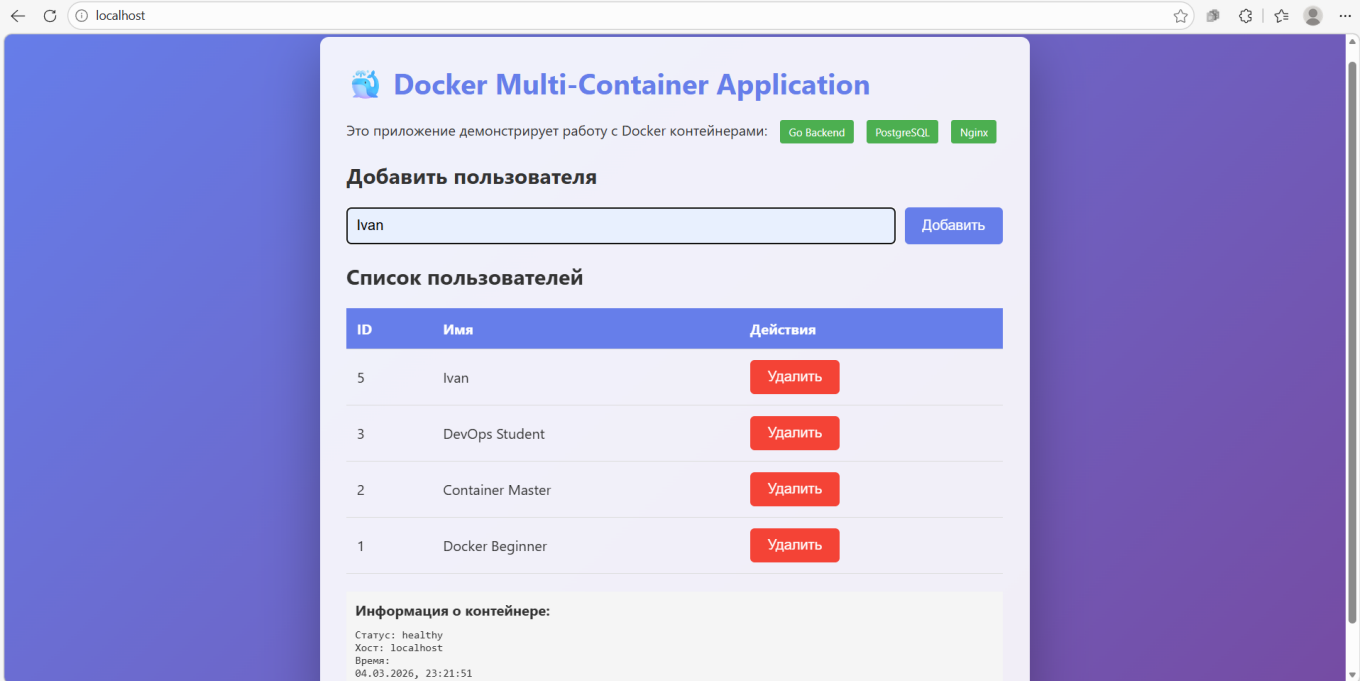
```
giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker network create -d bridge my-new-network
9662fae7a6356e424dcafe03e4c39faf5962680cfbc088478163bb2346ee9803

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker run -d --name test-postgres --network my-new-network -e POSTGRES_PASSWORD=supersecret -e POSTGRES_DB=mynewapp postgres:15-alpine
a710d1022296d84fac7bd0235e0e00d2b814347c0e992738c0c9de4e5f1318df

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker run -d --name newapp --network my-new-network -p 8088:80 nginx:alpine
1ecce690da38a30b111d76a1047bcc5ed3564ccbaef55966311345bc85ca4d93

giper@LAPTOP-31PBTHTU MINGW64 /c/dev/containers-lab-1 (main)
$ docker exec newapp ping test-postgres
PING test-postgres (172.19.0.2): 56 data bytes
64 bytes from 172.19.0.2: seq=0 ttl=64 time=0.594 ms
64 bytes from 172.19.0.2: seq=1 ttl=64 time=0.250 ms
64 bytes from 172.19.0.2: seq=2 ttl=64 time=0.308 ms
64 bytes from 172.19.0.2: seq=3 ttl=64 time=0.182 ms
64 bytes from 172.19.0.2: seq=4 ttl=64 time=0.184 ms
```

2. Скриншоты работающего приложения

2.1 Главная страница



2.2 Добавление пользователя

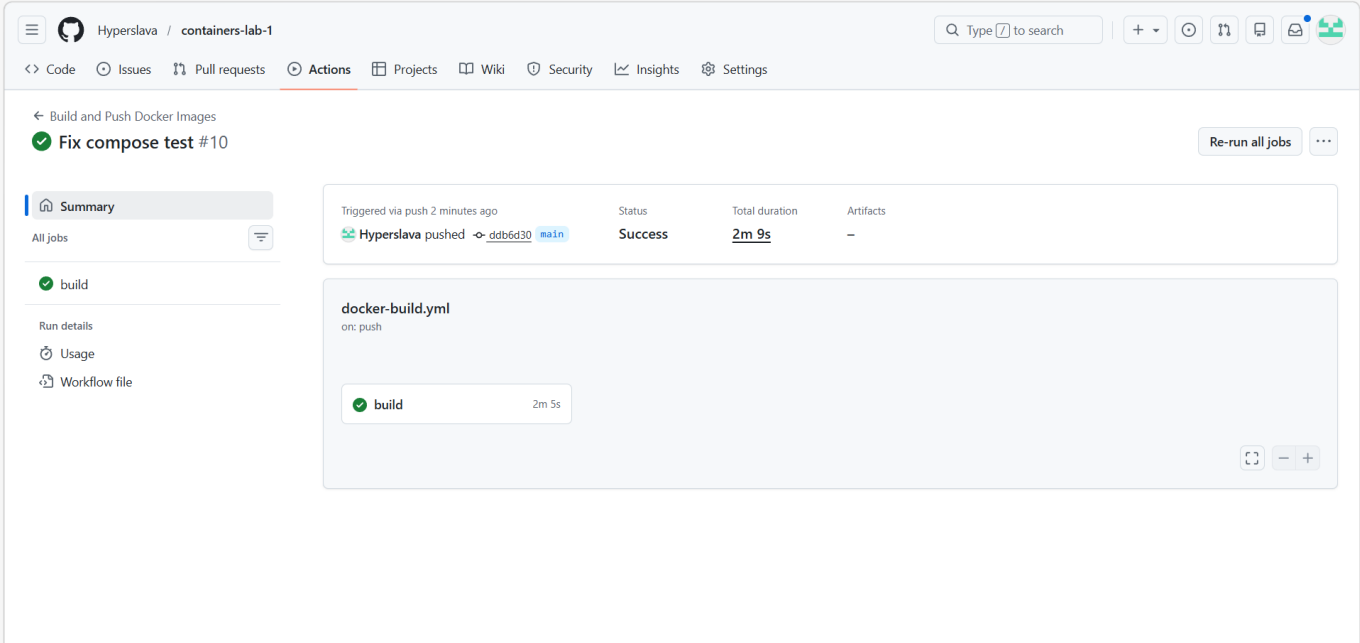


2.3 Список пользователей в БД

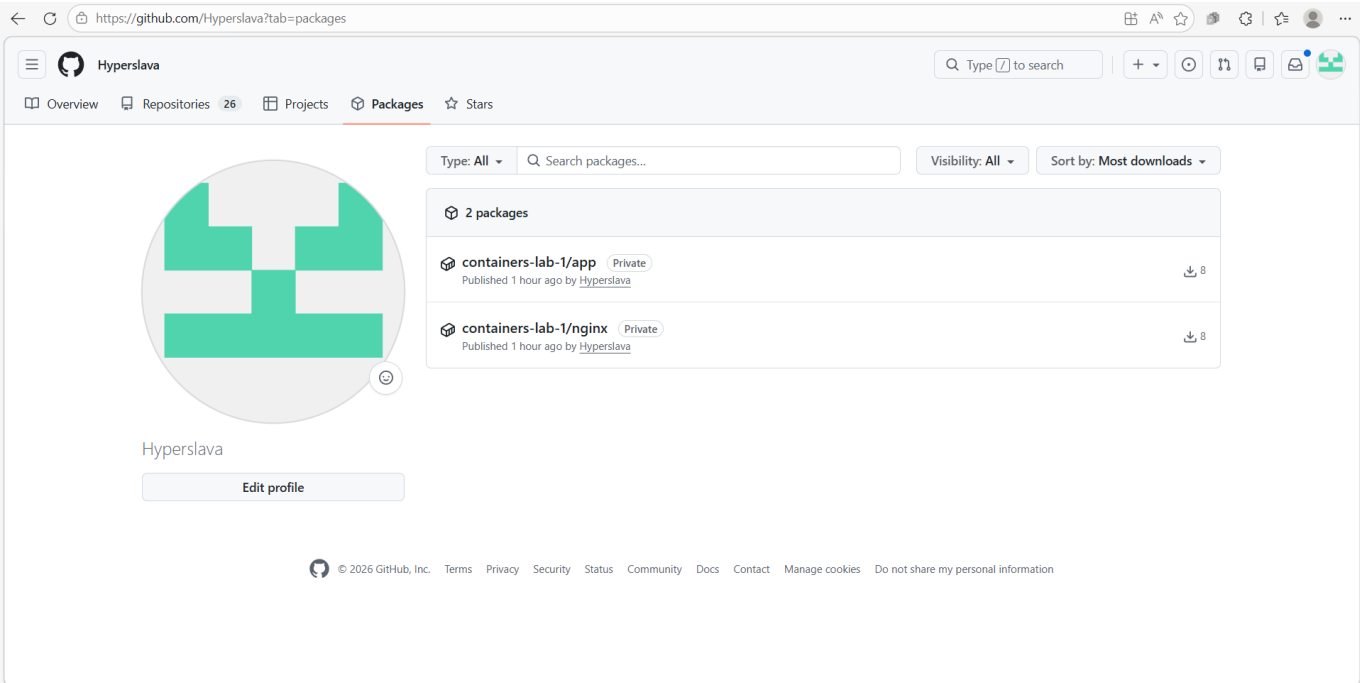
```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (main)
$ docker exec -it lab1-postgres psql -U postgres -d myapp -c "SELECT * FROM users;"
 id |       name       |         created_at
----+------------------+----------------------------
  1 | Docker Beginner  | 2026-03-04 18:41:41.965928
  2 | Container Master | 2026-03-04 18:41:41.965928
  3 | DevOps Student   | 2026-03-04 18:41:41.965928
  5 | Ivan             | 2026-03-04 20:21:34.552928
(4 rows)


What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug lab1-postgres
    Learn more at https://docs.docker.com/go/debug-cli/

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (main)
```

3. GitHub Actions

3.1 Успешный запуск workflow



3.2 Опубликованные образы в GHCR



4. Выводы

Из нового я узнал, как создать веб-приложение с бэкендом на Go, как проверять связь между контейнерами, как выкладывать контейнеры в GitHub container registry. Самое трудное было разобраться с запуском workflow и выкладыванием контейнеров в GitHub container registry.